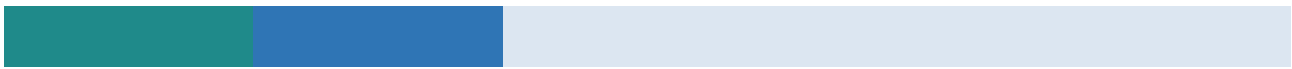


SQL Assessment Practice Questions

25 progressive challenges from fundamentals to interview-level SQL



SKILLS COVERED

- Filtering and aggregation
- JOINS and self joins
- Subqueries and CTE thinking
- Window functions
- Real assessment patterns

Designed as a question-only workbook

LEVEL 0 How to use this workbook

Work sequentially. Write and test each query before moving to the next level.

1	Solve Write the query independently before checking notes or earlier examples.
2	Test Validate the result against the sample data, including NULLs and duplicate values.
3	Explain Be able to describe why the JOIN, grouping, subquery or window function works.
4	Refine Improve aliases, readability and handling of edge cases.

Progress map

Stage	Focus	Questions
Level 1	Basic	Q1-Q5
Level 2	JOIN + Aggregation	Q6-Q10
Level 3	Subqueries	Q11-Q13
Level 4	Window Functions	Q14-Q17
Level 5	Real Assessment Questions	Q18-Q25

LEVEL 1 Basic

Build confidence with filtering, scalar aggregates, subqueries and grouped summaries.

Q1**Find all employees whose salary is greater than 70,000.****WHERE****Q2****Find the highest salary in the employees table.****MAX****Q3****Find the second-highest salary. Do not use LIMIT 1 OFFSET 1.***Hint: Solve it using a subquery.***SUBQUERY MAX****Q4****Find the average salary of employees in each department.****GROUP BY AVG****Q5****Find departments where the average salary is greater than 60,000.****GROUP BY HAVING**

LEVEL 2 JOIN + Aggregation

Combine related tables while preserving the correct population and calculating customer-level measures.

Q6

Find the number of orders placed by each customer. Include customers who placed zero orders.

Hint: Think carefully about which JOIN preserves every customer.

LEFT JOIN COUNT

Q7

Find customers who placed more than 2 orders.

JOIN GROUP BY HAVING

Q8

Find the total amount spent by each customer. Include customers who have placed no orders.

LEFT JOIN SUM COALESCE

Q9

Find the customer who spent the highest total amount.

JOIN SUM ORDERING

Q10

Find customers who never placed an order.

Hint: This is a common assessment pattern.

ANTI JOIN

LEVEL 3 Subqueries

Compare each row with a calculated benchmark at organisation or department level.

Q11

Find employees whose salary is greater than the overall average salary.

SUBQUERY AVG

Q12

Find employees whose salary is greater than their department's average salary.

Hint: Compare each employee's salary with the average for that employee's department.

CORRELATED SUBQUERY

Q13

Find the third-highest salary without manually specifying a salary value.

SUBQUERY DISTINCT

LEVEL 4 Window Functions

Use ranking functions to compare rows while retaining row-level detail.

Q14

Rank employees by salary from highest to lowest.

Hint: Output: employee_name | salary | salary_rank. Use RANK().

RANK

Q15

Find the highest-paid employee in each department.

Hint: Use RANK() with PARTITION BY.

PARTITION BY RANK

Q16

Find the top 2 highest-paid employees in each department.

PARTITION BY RANK

Q17

Find employees who have the same salary.

Hint: Inspect the data carefully. At least one salary is duplicated.

COUNT WINDOW

LEVEL 5 Real Assessment Questions

Apply multiple techniques together and explain your choice of approach.

Q18

Find the top-selling product in each category based on total sales amount.

Hint: JOIN → GROUP BY → SUM → RANK or ROW_NUMBER → PARTITION BY.

JOIN WINDOW

Q19

Find customers whose total spending is greater than the average customer spending.

Hint: Customer → total spending → average of customer totals.

CTE AGGREGATION

Q20

Find employees whose salary is greater than their manager's salary.

Hint: Use a self join.

SELF JOIN

Q21

Find customers who haven't placed any orders. Solve it two ways.

Hint: 1. LEFT JOIN 2. NOT EXISTS

LEFT JOIN NOT EXISTS

Q22

Find total revenue for each product category.

Hint: Expected output: category | total_revenue

JOIN SUM

LEVEL 5 Real Assessment Questions, continued

Finish with date aggregation, duplicate detection and departmental ranking.

Q23

Find total sales for each month.

Hint: Expected output: month | total_sales. Extract the month from order_date.

DATE GROUP BY

Q24

Find the salaries that occur more than once.

Hint: Expected output: salary. The data contains 75,000 more than once. Use GROUP BY with HAVING COUNT() > 1.*

GROUP BY HAVING

Q25

Find the second-highest salary in each department.

Hint: Combine PARTITION BY with RANK or DENSE_RANK, then filter the ranked result.

DENSE_RANK PARTITION BY

Completion checklist

- ☐ Query returns the requested columns only
- ☐ JOIN type matches the inclusion requirement
- ☐ Aggregates use the correct grouping level
- ☐ NULLs and duplicate values are handled deliberately
- ☐ Aliases are meaningful and the query is readable
- ☐ You can explain the logic aloud